Introduction and data:

Artificial Intelligence has been growing at an exponential rate and is being incorporated into daily life in such a way that humans may need to interact with it daily. This needs trust between humans and AI. Whether humans are trusting AI generated advice and how this trust compares to their trust in human experts is an ongoing research question. A research done by Dietvorst et al., in 2015 has the results that people often choose human forecasters over algorithmic forecasters even after they see them outperform humans.

The dataset used for this analysis, to understand the trust between human and AI when making decisions is called the Human - AI interactions dataset created by Kailas Vuppalapati and others, this dataset can be found in Github. It contains over 30000 Human-AI interactions. They were collected as part of the study on how humans make decisions with or without advice. They recruited over 1100 crowd workers and expert dermatologists and then asked them to perform a variety of binary classification tasks. Then each participant provided two responses, first an initial judgement based on just the presented information, and then a revised judgement after receiving advice that was labelled as either given from a human or an AI algorithm. This focuses on how individuals make classification decisions both after and before receiving advice. One important factor here is that the advice is said to be coming from either a human or AI and checking how that influences the decision.

My research question is, how does the source of advice influence participants' confidence and accuracy in decision-making tasks?
Null hypothesis is, the source of advice does not affect participant's confidence and accuracy in decision making tasks. Alternate hypothesis is participant's confidence and accuracy in decision making tasks differs based on whether the advice comes from human or AI and is higher if the advice is from human.

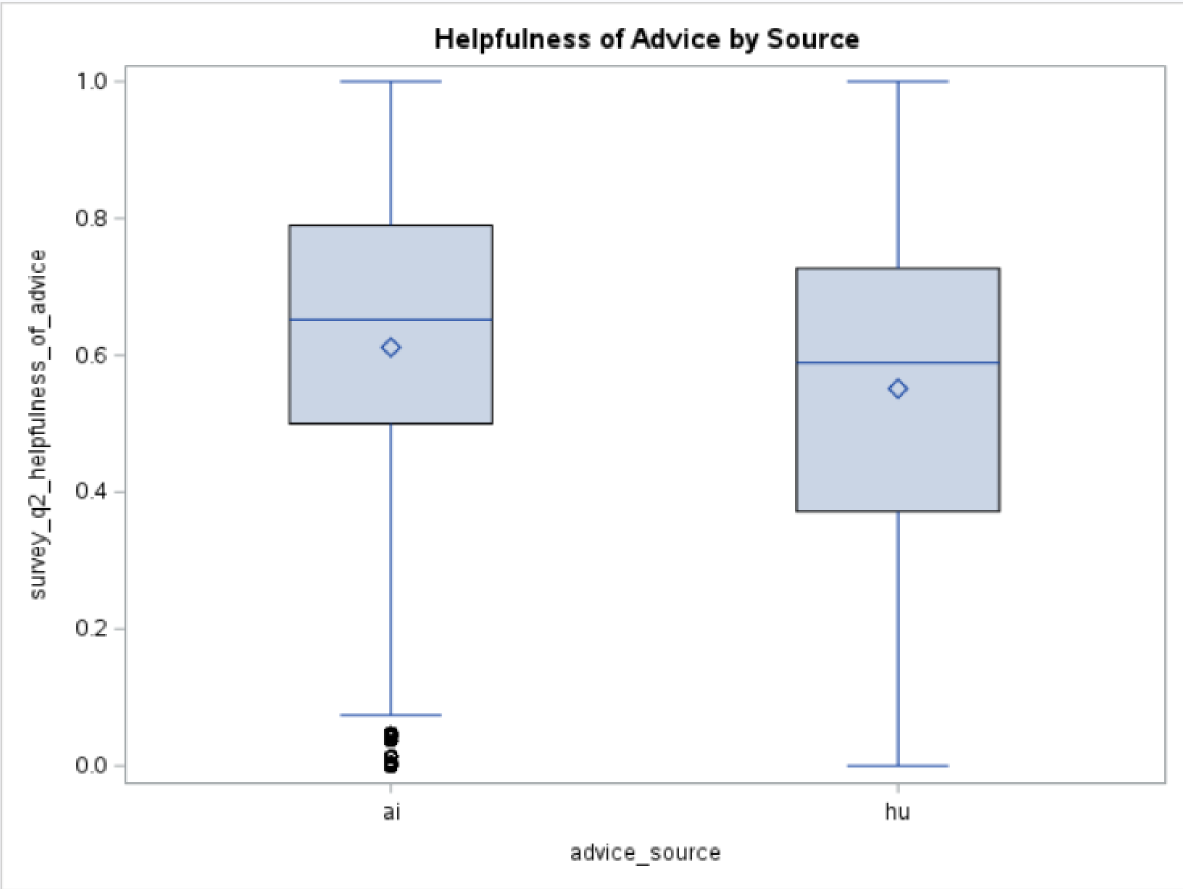
In this study, the primary outcome variables were trust in advice and changes in decision accuracy after receiving advice. Trust was measured using a continuous survey item (survey_q3_trust_in_advice, ranging from 0–1), while decision improvement was captured through a derived variable (response_change = response_2 – response_1), where positive values indicate improved accuracy after advice. The central predictor of interest was the perceived source of advice (advice_source), which was randomly assigned to either human or AI, allowing us to directly test whether people trust one source more than the other. We also modeled the direction and correctness of the advice itself through the advice variable (−1 to

1), indicating whether the advice pointed toward the correct answer. This ensured that our analysis accounted for whether participants were guided accurately or misled.

To explain individual differences in how advice was used, we incorporated demographic and experience-based covariates. Gender (male, female, other/unknown), age, and education level (1–8 scale) were included to account for variation in experience, training, and decision-making style. Socioeconomic status was reported on a 1–10 scale, capturing participants perceived financial and social standing. These measures were included because both access to technology and comfort with AI systems may vary across education and socioeconomic groups. We also included programming experience (True/False) to account for familiarity with computing, and AI use in daily life (survey_q5_AI_in_life, ranging from 0–1) to capture exposure to AI tools outside the experiment. Finally, we accounted for the type of task (task_name) being completed (e.g., art, cities, sarcasm, census, dermatology), since differences in domain knowledge or difficulty could influence both confidence and accuracy.

Together, these variables allowed us to evaluate whether the perceived source of advice influences trust and decision improvement, while controlling for task difficulty, advice correctness, and participant background characteristics.

The dataset contained 35,670 observations, roughly evenly split between advice labeled as AI (50.39%) and human (49.61%), ensuring a balanced experimental design. Gender was relatively balanced, with 51% identifying as female and 46% as male. A majority of participants reported no programming experience (60%), while 35% reported some experience, suggesting that most respondents were not technically trained. Participants completed one of five task types, with art and sarcasm tasks being the most common, and dermatology the least common. Performance outcomes varied following advice: 36% of cases improved, 22% worsened, and 41% showed no change, with slightly higher improvement rates when advice was labeled as coming from AI. These patterns highlight variability in how advice is incorporated and motivate further modeling to understand which factors influence trust and performance.



Helpfulness of Advice by Source
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survey_q2_helpfulness_of_advice
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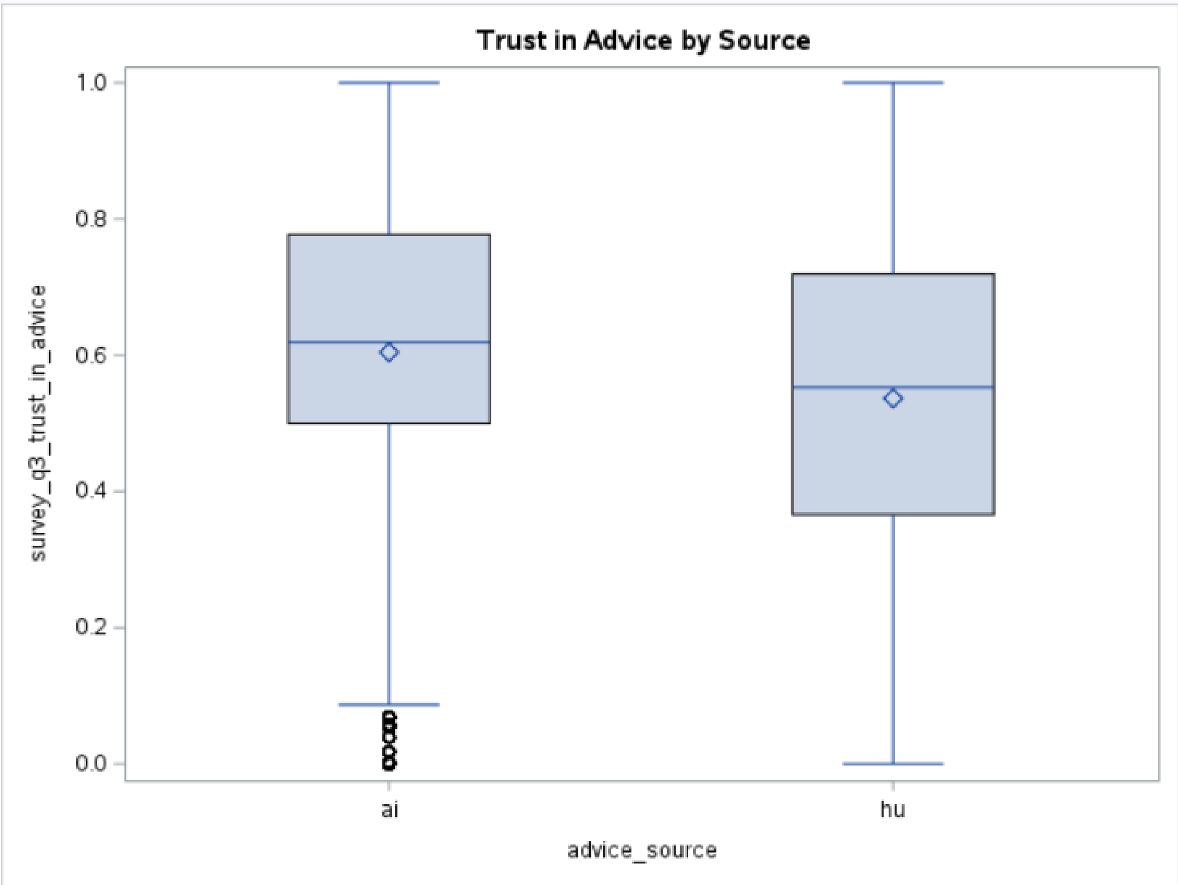

Trust in Advice by Source
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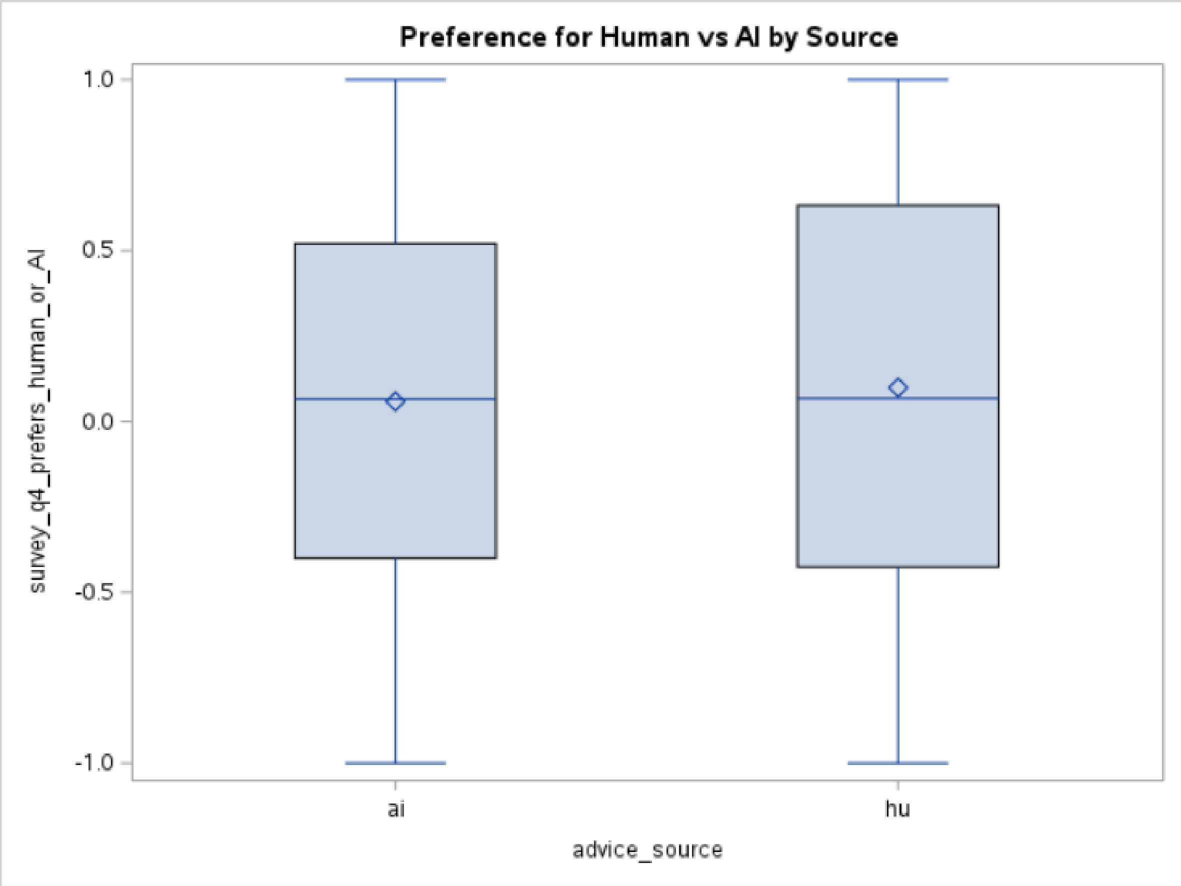
Preference for Human vs AI by Source
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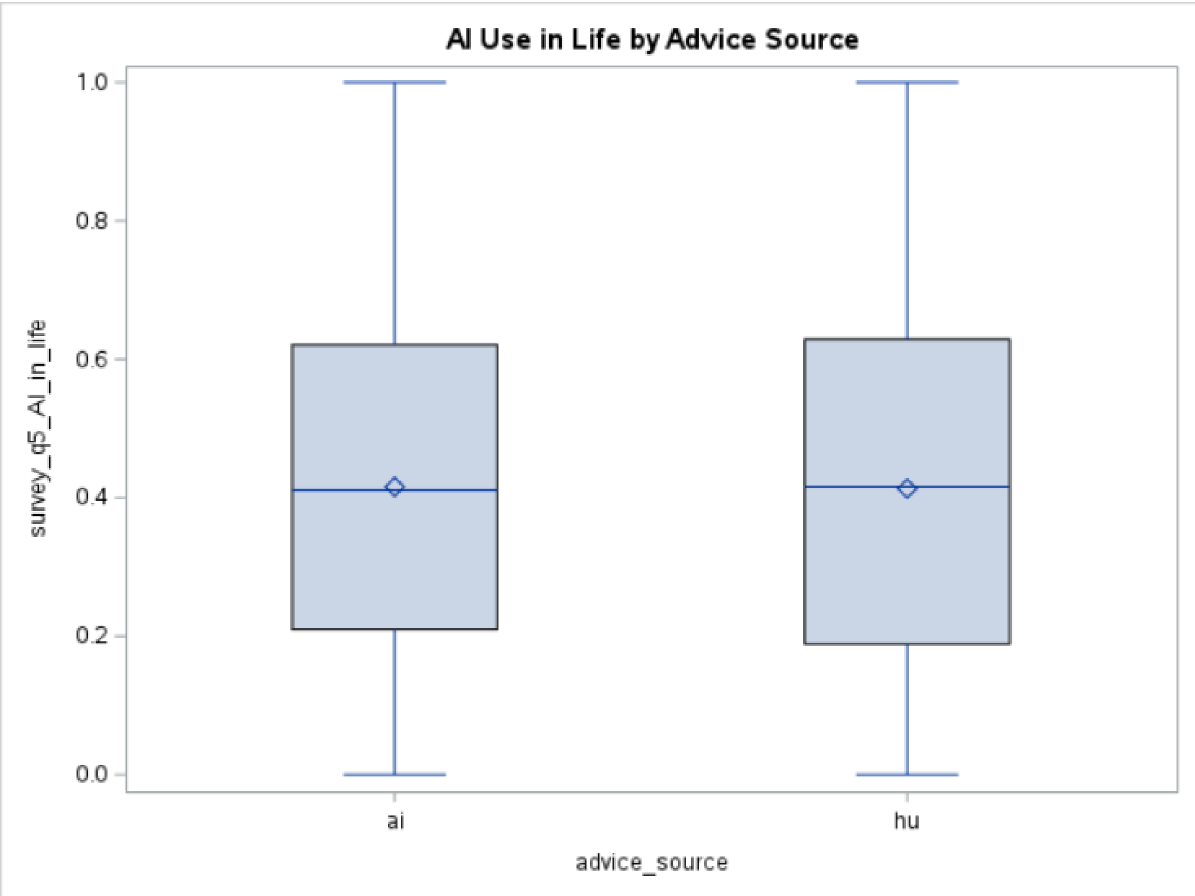
AI Use in Life by Advice Source
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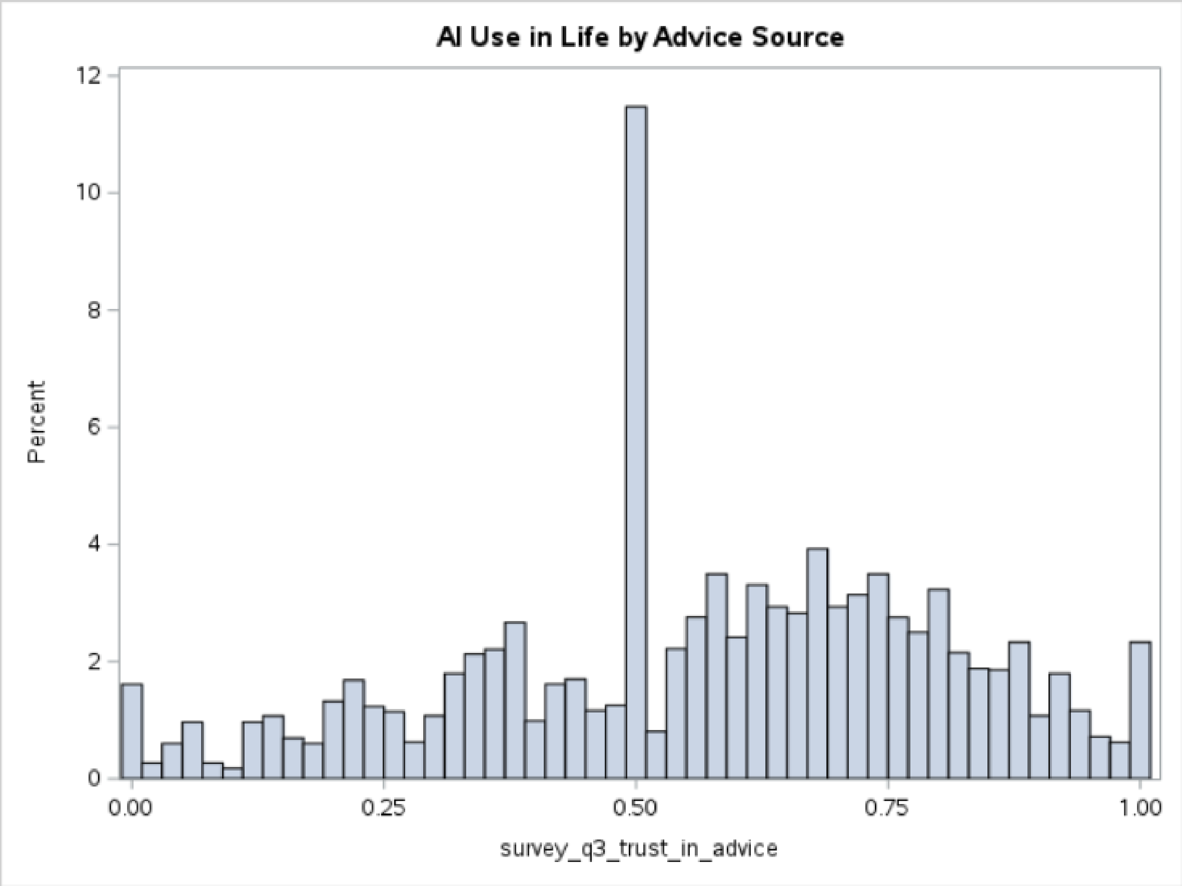
AI Use in Life by Advice Source
Percent
survey_q3_trust_in_advice
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8
10
12
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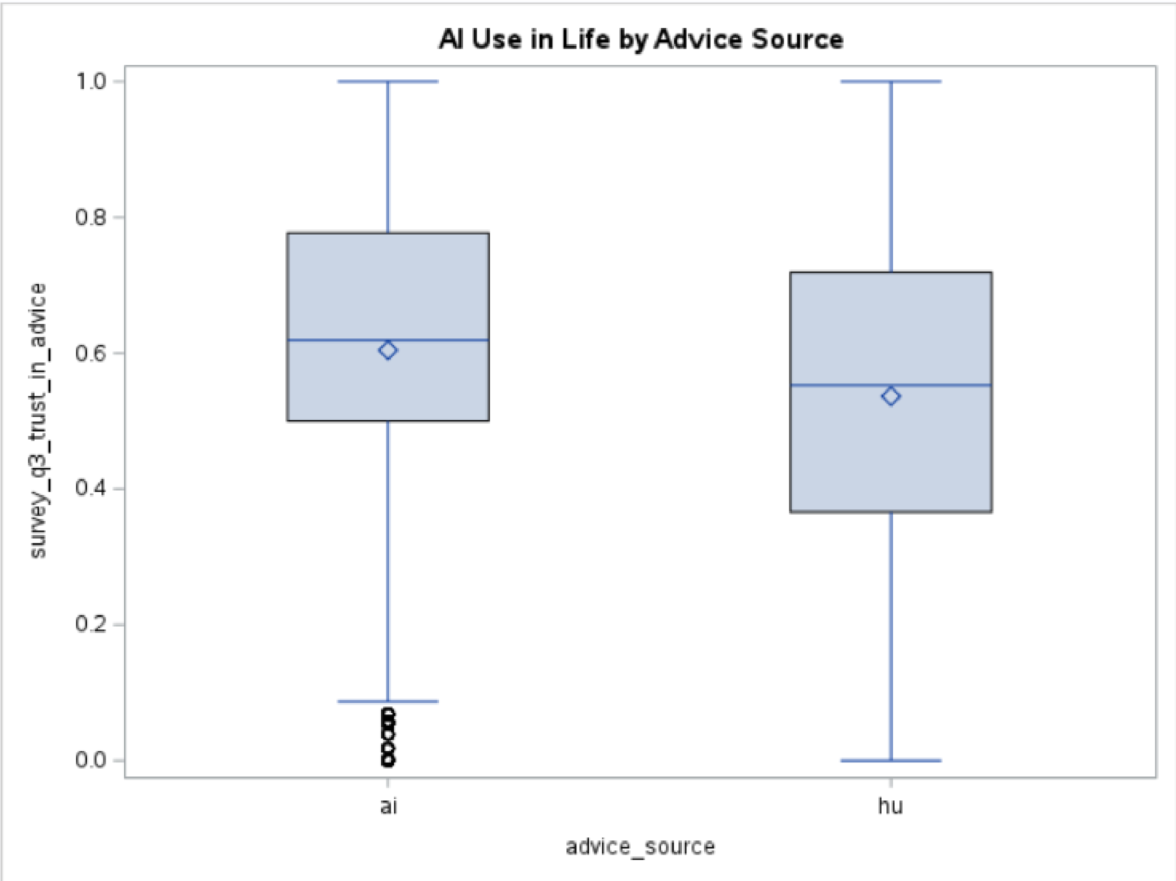
0.50
0.75
1.00


AI Use in Life by Advice Source
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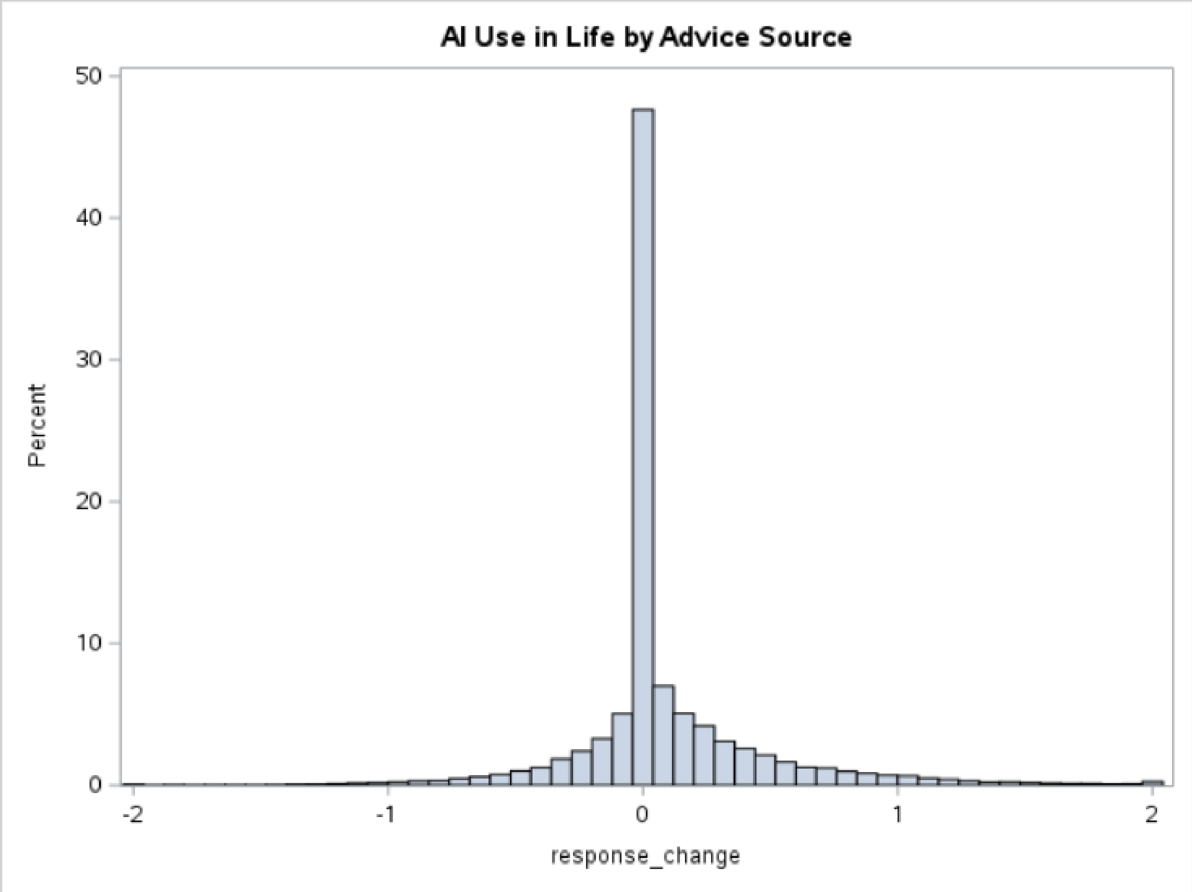
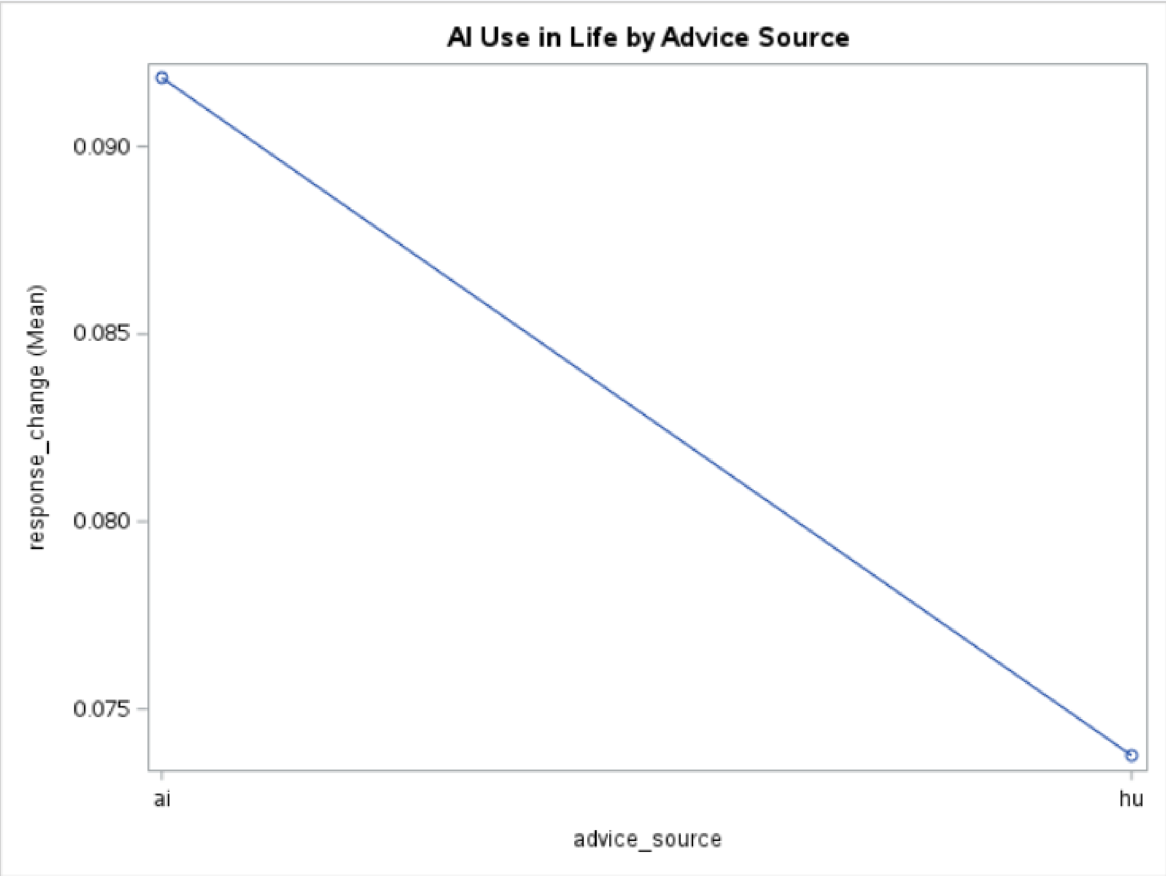


Across all visualizations, participants generally rated the advice as moderately helpful and trustworthy, with typical scores above the midpoint of the scale. Helpfulness and trust were slightly higher for advice labeled as AI, not consistent with the idea that people tend to favor human expertise, but the interquartile ranges for human and AI substantially overlapped. This overlap indicates that many participants responded similarly to advice regardless of the source. Preference ratings for whether humans or AI perform better were centered close to zero for both conditions, showing no strong overall preference, although the wide spread suggests that individual beliefs varied considerably. Likewise, participants' reported use of AI in daily life was very similar across advice conditions, confirming that random assignment worked and that no systematic differences existed between the two groups before advice was given. Histogram results also indicated that trust ratings were frequently clustered near the midpoint, with some participants reporting either extremely high or low trust. Finally, the distribution of response changes showed that a large proportion of participants did not change their decision after receiving advice, while others showed small improvements or declines. Overall, the EDA suggests that advice was generally viewed positively, and although participants expressed slightly higher trust and helpfulness for human advice, their behavioral responses were broadly similar across sources.



This line plot shows the mean response change for AI versus human advice. On average, participants improved slightly more when the advice was labeled as AI. However, the difference between the two conditions is small, suggesting that both sources led to similar performance improvements, even though participants reported somewhat higher trust in human advice.

Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
age	34783	31.30693	11.66951	1088949	18.00000	81.00000
education	35670	5.49493	1.38677	196004	1.00000	8.00000
socioeconomic_status	34751	5.25087	1.62418	182473	1.00000	10.00000
survey_q2_helpfulness_of_advice	35670	0.58172	0.25319	20750	0	1.00000
survey_q5_AI_in_life	35670	0.41458	0.26663	14788	0	1.00000

Table 1 reports summary statistics for the key demographic and attitudinal variables used in the analysis. Age ranged from 18 to 81 years, with a mean of 31.3, indicating a relatively young participant population. Education levels ranged from 1 to 8, representing categories from secondary education through postgraduate degrees, with a mean around 5.5, suggesting that many participants had at least some higher education. Socioeconomic status also exhibited a broad range (1–10), with a mean of 5.25, indicating a mid-level distribution. Both attitudinal variables had means above the midpoint of the scale: perceived helpfulness of advice averaged 0.58, and reported AI use in daily life averaged 0.41, suggesting moderately positive attitudes toward advice and modest exposure to AI tools.

Pearson Correlation Coefficients
Prob > |r| under H0: Rho=0
Number of Observations

	age	education	socioeconomic_status	survey_q2_helpfulness_of_advice	survey_q5_AI_in_life
age	1.00000 34783	0.20938 <.0001 34783	0.02588 <.0001 34751	-0.09163 <.0001 34783	-0.13967 <.0001 34783
education	0.20938 <.0001 34783	1.00000 35670	0.30178 <.0001 34751	-0.02508 <.0001 35670	0.04797 <.0001 35670
socioeconomic_status	0.02588 <.0001 34751	0.30178 <.0001 34751	1.00000 34751	0.06921 <.0001 34751	0.01871 0.0005 34751
survey_q2_helpfulness_of_advice	-0.09163 <.0001 34783	-0.02508 <.0001 35670	0.06921 <.0001 34751	1.00000 35670	0.19561 <.0001 35670
survey_q5_AI_in_life	-0.13967 <.0001 34783	0.04797 <.0001 35670	0.01871 0.0005 34751	0.19561 <.0001 35670	1.00000 35670

The correlation matrix shows that most variables were only weakly related to one another. The strongest correlation was between education and socioeconomic status ($r \approx 0.30$), which is conceptually reasonable. All other associations were below $r = 0.21$. Importantly, no pair of variables demonstrated high correlations, and the absence of strong linear relationships indicates no concern for multicollinearity among the predictors. This means that including age, education, socioeconomic status, and survey variables in the same model will not inflate standard errors or distort coefficient estimates. These results justify treating the demographic and attitudinal measures as separate control variables in the modeling process.

Methodology:

We divided our analysis into two complementary models because the research question concerns both how participants feel about the advice and how they behave after receiving it. The confidence model focuses on self-reported trust in the advice, treating survey_q3_trust_in_advice as the outcome and asking whether participants say they trust AI-labeled advice more or less than human-labeled advice, after accounting for individual characteristics and task type. The accuracy model, in contrast, uses response_change as the outcome and examines whether advice from AI or humans actually moves participants' responses closer to the correct label. This separation is important because trust and performance do not necessarily align, participants may report higher or lower confidence in one source while showing similar or even better accuracy changes with the other. By fitting both models, we are able to compare perceived trust and actual decision improvement side by side and obtain a more complete picture of how advice source influences human decision making. To investigate how the source of advice relates to participants' confidence, we modeled trust in advice (survey_q3_trust_in_advice, scaled 0–1) as a continuous outcome using a general linear model (GLM). A linear model is appropriate here because we are interested in how average trust scores vary across advice conditions and participant characteristics.

We began with a simple model including only advice source (AI vs human) as a categorical predictor, using human as the reference group. This provided an initial estimate of the mean difference in trust between AI-labeled and human-labeled advice. We then added task type (task_name) to control for differences across the five task domains (art, cities, sarcasm, census, dermatology), since task content could independently affect confidence.

Next, we considered a broader set of predictors to account for individual differences. Using proc glmselect with stepwise selection based on AIC, we evaluated potential main effects of age, education, socioeconomic status, perceived accuracy of advice, gender, programming experience, and task type, along with advice source. This procedure identified a parsimonious main-effects model that retained predictors that improved model fit without overfitting. We then refit this selected model with proc glm and obtained coefficient estimates and fitted values for interpretation and diagnostics.

Because our research question also asked whether the effect of advice source might depend on context, we fit an additional model including interaction terms between advice source and perceived accuracy, advice source and task type, and advice source and programming experience. These interactions allowed us to test whether differences in trust between AI and human advice were stronger or weaker for certain tasks, stated accuracy levels, or participant experience profiles.

For this model, we also checked the usual linear model assumptions. We generated a residuals-versus-fitted plot to assess linearity and constant variance and used histograms and Q–Q plots of the residuals from the main-effects model to evaluate the normality assumption.

The residuals were approximately centered around zero with no strong patterns in the residual–fit plot, suggesting that the linear functional form and constant variance were reasonable. The normality assumption was moderately violated, with some skew and heavier tails, which is expected with a bounded 0–1 outcome and a large sample size. However, based on course guidance and the large N, we judged that the general linear model was still appropriate for estimating average effects. Pairwise correlations among predictors were modest, indicating no serious multicollinearity. Because there were no severe violations, we did not apply transformations or remove predictors and proceeded with the final GLM for trust in advice.

For the accuracy model, we examined how advice influenced changes in participants' decisions by using the variable response_change, defined as response_2 – response_1. Both response_1 and response_2 are continuous measures ranging from −1 to 1, where values greater than 0 indicate that the participant selected the correct label and values less than 0 indicate selection of the incorrect label, with the magnitude reflecting how strongly the response aligns with the correct class. As a result, response_change is a continuous measure that can take values roughly between −2 and 2. Positive values indicate that the participant's final response moved closer to the correct side after receiving advice (an improvement), negative values indicate movement toward the incorrect side (a deterioration), and values near zero correspond to little or no change in accuracy. Because we were interested in how the average change in accuracy depends on the advice source and participant characteristics, we modeled response_change using a general linear model.

We began with a simple model that included only the perceived source of advice (AI versus human, using human as the reference category) as a predictor. This provided an initial estimate of the mean difference in accuracy change between AI-labeled and human-labeled advice. We then considered a richer set of predictors to account for the content and context of the advice and for individual differences. Using a stepwise selection procedure based on AIC, we evaluated main effects of advice source, the advice value (advice, indicating whether the advice pointed toward the correct or incorrect label), perceived accuracy of the advice, age, education, socioeconomic status, gender, programming experience, and task type. The stepwise procedure selected a parsimonious subset of these predictors that improved model fit without making the model unnecessarily complex. Based on this selection, we refit a main-effects model that included advice source, advice, age, education, programming experience, and task type, and obtained coefficient estimates along with fitted values and residuals for further assessment.

Because our research question also asked whether the effect of advice source might depend on whether the advice itself was helpful or misleading, we fit an additional model that included an interaction term between advice source and advice (advice_source*advice), along with age, education, programming experience, and task type. This interaction allowed us to test whether the difference between AI and human advice was larger when the advice pointed toward the correct label than when it pointed toward the incorrect label. Model assumptions were evaluated using residual diagnostics. We examined a residuals-versus-fitted plot to assess linearity and constant variance and used histograms and Q–Q plots of the residuals from the main-effects model to assess normality. Residuals were centered around zero with no strong systematic pattern in the residual–fit plot, suggesting that the linear model provided a reasonable approximation. Although the residual distribution showed some deviation from perfect normality, which is expected given the bounded and somewhat discrete nature of response_change, the large sample size makes the general linear model adequate for summarizing average differences in accuracy change. Correlations among predictors were modest, so there was no indication of problematic multicollinearity, and no transformations or exclusions were required.

Results:

Confidence Model:

We first fit a baseline model with advice source as the only predictor of trust (survey_q3_trust_in_advice). This model showed a highly significant effect of advice source on trust, $F(1{,}35668) = 749.02, p < .0001$, but explained only about 2.1% of the variance in trust ($R^2 = 0.0206$). On average, participants reported higher trust when the advice was labeled as coming from AI rather than from a human.

Adding task type to the model substantially improved fit. The model including advice source and task name explained about 5.6% of the variance ($R^2 = 0.0559$), with both predictors highly significant (advice source: $F = 777.8, p < .0001$; task type: $F = 334.2, p < .0001$). This indicates that trust levels varied across the different task domains in addition to the difference between AI and human advice.

We then used stepwise selection to build a richer main-effects model that included demographic and experience variables. The selected model contained advice source, age, education, socioeconomic status, perceived accuracy of the advice, gender, programming experience, and task type. This model explained about 7.5% of the variance in trust ($R^2 = 0.0754$)and fit significantly better than the simpler models, $F(12,34738) = 235.94, p < .0001$. In this model, advice source remained an important predictor: holding the other variables constant, trust for AI-labeled advice was on average 0.064 points higher than for human-labeled advice on the 0–1 scale (estimate = 0.0642, $t = 26.20, p < .0001$). Perceived accuracy was also a strong predictor: each one-unit increase in the perceived accuracy scale was associated with an increase of about 0.0048 in trust ($t = 25.61, p < .0001$), and socioeconomic status and education were positively related to trust as well. Age showed a small negative association (older participants reported slightly lower trust), and there were modest differences by gender and programming-experience categories. Task type remained highly significant, indicating that some tasks elicited systematically higher or lower trust even after adjusting for individual characteristics.

Finally, we estimated an extended model including interactions between advice source and perceived accuracy, task type, and programming experience. This model modestly increased the explained variance to about 8.5% ($R^2 = 0.0846$). The interaction terms were jointly significant (e.g., advice source × task type: $F = 58.05, p < .0001$; advice source × programming experience: $F = 60.05, p < .0001$), indicating that the difference in trust between AI and human advice was not constant across all tasks and experience levels. In this interaction model the overall main effect of advice source was no longer significant on its own, $F = 1.09, p = .2955$, which is consistent with the interpretation that the effect of advice source depends on context: for some combinations of task and participant characteristics the trust difference between AI and human advice is larger, while for others it is minimal. Overall, the confidence model shows that participants' trust is driven primarily by how accurate and credible the advice is perceived to be, with AI labels often enjoying a small trust advantage, but with meaningful variation across tasks and subgroups.

Accuracy Model:

We first examined whether accuracy change differed by advice source alone. In a baseline model with response_change regressed only on advice source, the effect of advice source was statistically significant, $F(1,35668) = 20.22, p < .0001$, but the model explained less than one tenth of one percent of the variance in response change ($R^2 = 0.0006$). Mean response change was positive overall (mean ≈ 0.083), indicating a small tendency for answers to move toward the correct label after advice, and this improvement was slightly larger for AI-labeled advice than for human-labeled advice.

Adding task type to the model increased the explained variance to about $R^2 = 0.0044$. Both advice source and task name were significant predictors in this two-factor model (advice source: $F = 20.30, p < .0001$; task name: $F = 34.65, p < .0001$), suggesting that the extent of accuracy change varied somewhat across domains in addition to the small difference between AI and human advice.

A more comprehensive model was then built using stepwise selection. The final selected model included advice source, the advice value (advice), age, education, programming experience, and task type. This model explained about 10.2% of the variance in response change ($R^2 = 0.1016$)and provided a highly significant overall fit, $F(9{,}34741) = 436.68, p < .0001$. Among the predictors, the advice value was by far the strongest: each one-unit increase in advice (on the −1 to 1 scale) was associated with an estimated 0.297 increase in response change ($t = 60.94, p < .0001$). This large effect confirms that when advice pointed more strongly toward the correct label, participants' final responses moved much more toward the correct side; when advice pointed toward the incorrect label, response change tended to be negative.

Even after controlling for advice value and the other covariates, advice source remained a significant predictor. AI-labeled advice was associated with an additional 0.018 increase in response change compared to human-labeled advice ($t = 4.72, p < .0001$), a small but detectable improvement relative to the overall mean change of about 0.083. Several participant characteristics also showed significant, but comparatively modest, associations with accuracy change. Older participants exhibited slightly smaller improvements (age estimate = −0.00094, $t = -5.36, p < .0001$), while higher education was associated with somewhat larger improvements (estimate = 0.0037, $t = 2.46, p = .014$). Programming experience mattered as well: participants who reported no programming experience improved slightly less than those in the reference category (estimate = −0.0127, $t = -2.98, p = .0029$). Task type remained significant in the full model, with larger gains on the art and cities tasks (e.g., art estimate = 0.0546, $t = 11.74, p < .0001$) and smaller or even slightly negative changes for some other domains relative to sarcasm.

Overall, the accuracy model shows that what the advice says—whether it points toward the correct or incorrect label—is by far the dominant driver of changes in participants' answers, while the label of the source (AI vs human) has a smaller but still positive effect, with AI-labeled advice leading to slightly greater improvements in accuracy on average.

Discussion and Conclusion

This project set out to answer how the perceived source of advice, AI versus human, shapes participants' confidence in that advice and the extent to which their decisions become more accurate after receiving it. By separating the analysis into a confidence model (trust in advice) and an accuracy model (change in response), we were able to compare what people say they feel about advice with how they actually behave when they use it.

The confidence model showed that the perceived source of advice does matter, but in a way that contrasts with older findings suggesting people systematically distrust algorithms. After controlling for task, perceived accuracy, and demographic factors, AI-labeled advice was associated with a small but reliable increase in trust relative to human-labeled advice. At the same time, perceived accuracy emerged as one of the strongest predictors of trust: participants reported substantially higher trust when they believed the advice was more accurate, regardless of whether it was labeled as human or AI. Task type and background characteristics such as socioeconomic status and education also contributed, but their effects were relatively modest. Interaction results further suggested that the trust gap between AI and human advice is not uniform; it varies across tasks and experience levels, indicating that context shapes how people react to AI labels.

The accuracy model provided a complementary picture focused on behavior rather than self-report. Here, the dominant factor was not the label on the advice but its content. The advice value, how strongly the advice pointed toward the correct versus incorrect label, had by far the largest effect on response change. When the advice more clearly supported the correct answer, participants' final responses moved closer to the correct side; when the advice pushed toward the wrong answer, their accuracy deteriorated. After accounting for this, advice source still had a small positive effect: AI-labeled advice produced slightly larger average improvements in accuracy than human-labeled advice, but the size of this effect was modest relative to the impact of advice correctness itself. Task type and demographic variables again played a secondary role, adding context but not overturning the main conclusion. Together, the two models suggest an important distinction: participants' trust and their performance are related but not identical. They appear willing to trust AI advice at least as much as, and sometimes more than, human advice, while their actual accuracy is driven mainly by whether the advice is correct.

There are several limitations that should be considered when interpreting these findings. First, although the dataset is large, the sample is not necessarily representative of the general population, and participants' prior exposure to AI or online studies may differ from that of other groups. Second, many of our key variables, such as trust, perceived helpfulness, and AI use in daily life, are measured on bounded Likert-type scales but were treated as numeric and approximately continuous in the linear models. This approach is common in practice and simplifies interpretation, but it assumes that the distances between scale points are equal and may underestimate nonlinearity or ceiling effects. Third, the main outcome variables, trust and perceived accuracy, are self-reported and may be influenced by demand effects or social desirability, while response_change is derived from a specific experimental design with bounded scales. Fourth, the linear models used here impose assumptions such as linear relationships and approximately normal residuals; diagnostic checks indicated some departures from normality, which we judged acceptable given the large sample size and our focus on average differences, but more flexible approaches (e.g., ordinal or generalized linear models) could better respect the measurement scale. Finally, the study focuses on perceived source labels rather than real differences in underlying systems; participants were told that advice came from a human or AI, but their reactions may depend on that framing rather than on actual system performance.

These limitations point directly to several directions for improvement and future work. Methodologically, future analyses could use mixed-effects models to more explicitly account for repeated observations within the same participant or task instance or explore generalized linear models that better respect the bounded nature of the outcomes. From a design perspective, it would be valuable to replicate this study in higher-stakes domains and with more diverse samples, as well as to compare conditions where the AI label is paired with transparent performance information, explanations, or uncertainty estimates. Another promising direction would be to study how trust evolves over time as participants repeatedly interact with the same AI system, rather than forming judgments after a single session. More broadly, future work could investigate how to calibrate trust so that it aligns with actual system performance, making sure that people neither over trust nor under trust AI relative to its real capabilities.

In summary, our analysis suggests that participants are not uniformly skeptical of AI advice. On the contrary, AI-labeled advice received slightly higher trust ratings and produced slightly larger accuracy gains than human-labeled advice, although these effects were relatively small compared to the influence of advice correctness and perceived accuracy. These results complicate the simple narrative that humans always prefer human experts over algorithms. Instead, they point toward a more nuanced picture in which labels, perceived accuracy, task context, and individual background jointly shape how people relate to AI, underscoring the importance of designing decision-support systems that are both reliable in their recommendations and calibrated to how users actually experience and use them.

Vodrahalli, K., Gerstenberg, T., & Zou, J. (2021). Do Humans Trust Advice More if it Comes from AI? An Analysis of Human-AI Interactions. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2107.07015

https://github.com/kailas-v/human-ai-interactions?tab=readme-ov-file